Password Hash Assignment Report

HASHCAT 6.2.6

KALI LINUX 4

MD5 HASH CRACKING

JOHN THE RIPPER PASSWORD CRACKING

Course	SWE 267P
Quarter	Spring 2026
Student	Zhenyu Song

This report documents the exploration of **password hashing techniques**, covering online and local hash generation/cracking, dictionary and brute-force attacks, and analysis of real-world leaked hash datasets.

Table of Contents

- Password Hash Assignment Report
  - Table of Contents
  - Part 1: Hash generation/cracking online
    - 1.1 Online Hash Generation
      - Free Online Services
    - 1.2 Local Verification (using md5sum)
    - 1.3 Online Hash Cracking
      - Free Online Services
    - 1.4 Analysis
  - Part 2: MD5 hash cracking locally on macOS using Hashcat
    - 2.1 Identify the Hash Type
    - 2.2 Crack hashes using Hashcat and Rockyou Wordlist
    - 2.3 Crack target hashes using Rockyou Wordlist + Standard Rules
    - 2.4 Crack target hashes using a curated and larger wordlist
    - 2.5 Summary for Part 2
  - Part 3: Passphrase cracking
    - 3.1 Crack passphrases using custom word list with custom rules
    - 3.2 Try adding another rule (Stacking Rules)
    - 3.3 Create a custom rule
    - 3.4 Summary for Part 3
  - Part 4: Document password cracking
    - 4.1 Extract Hash from LibreOffice Document (hashcat.odt)
    - 4.2 Crack Hash with Hashcat
    - 4.3 Summary for Part 4

Part 1: Hash generation/cracking online

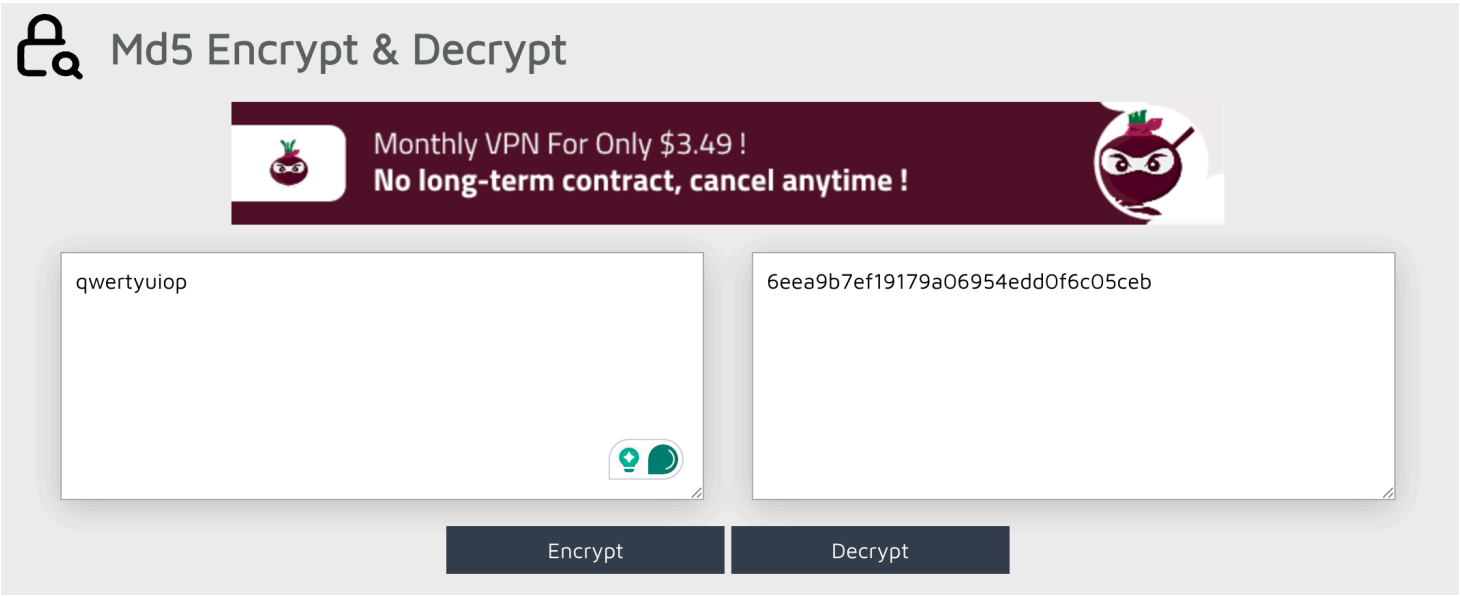
1.1 Online Hash Generation

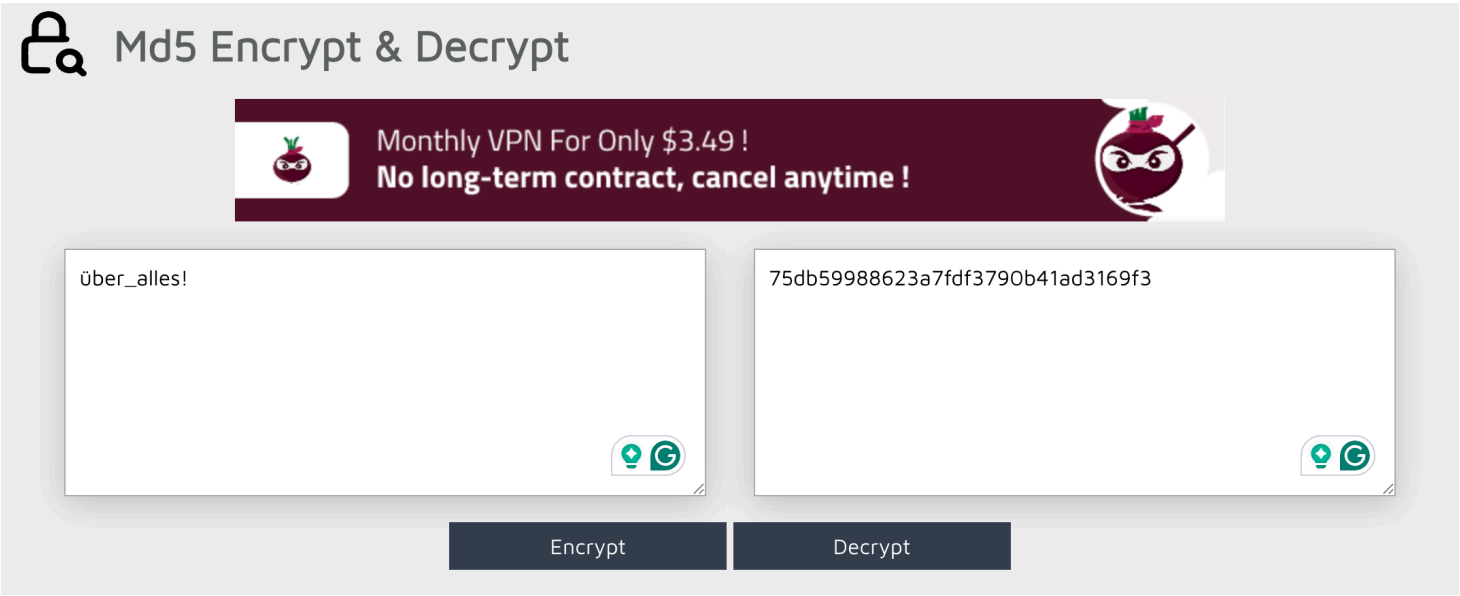
> Generate MD5 hashes using free online services.

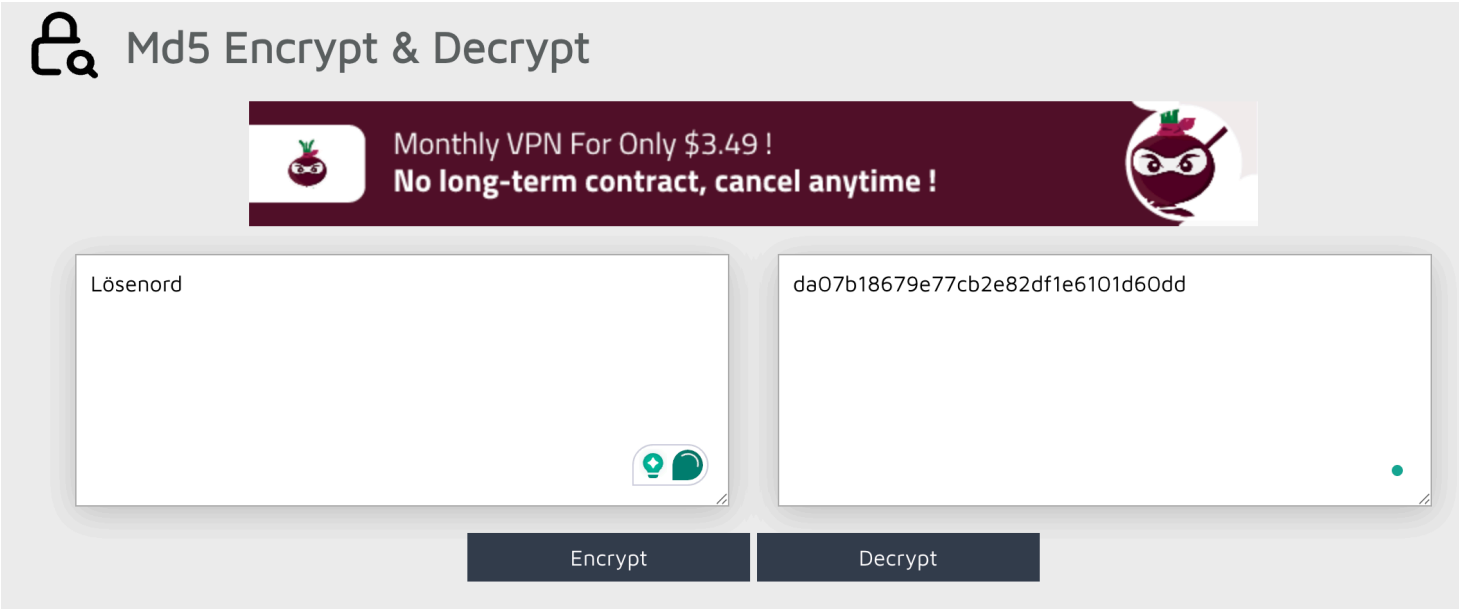
Free Online Services

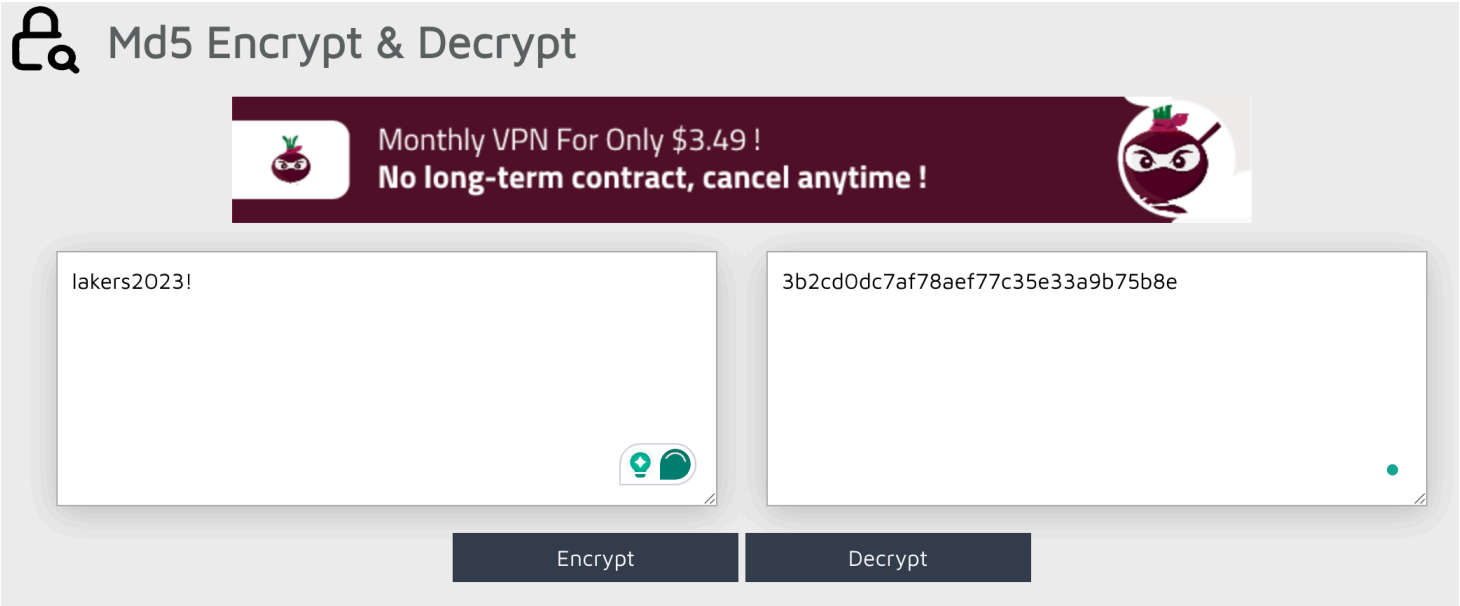
1. https://md5decrypt.net/en/
2. https://10015.io/tools/md5-encrypt-decrypt

Screenshots of https://md5decrypt.net/en/ Hash Generation:









Screenshots of https://10015.io/tools/md5-encrypt-decrypt Hash Generation:

MD5 Encrypt/Decrypt

Share | Add to Favs | Report Bug

Encrypter | Decrypter

Text

qwertyuiop

MD5 Hash

6eea9b7ef19179a06954edd0f6c05ceb

Encrypt › | Reset | Copy

MD5 Encrypt/Decrypt

Share | Add to Favs | Report Bug

Encrypter | Decrypter

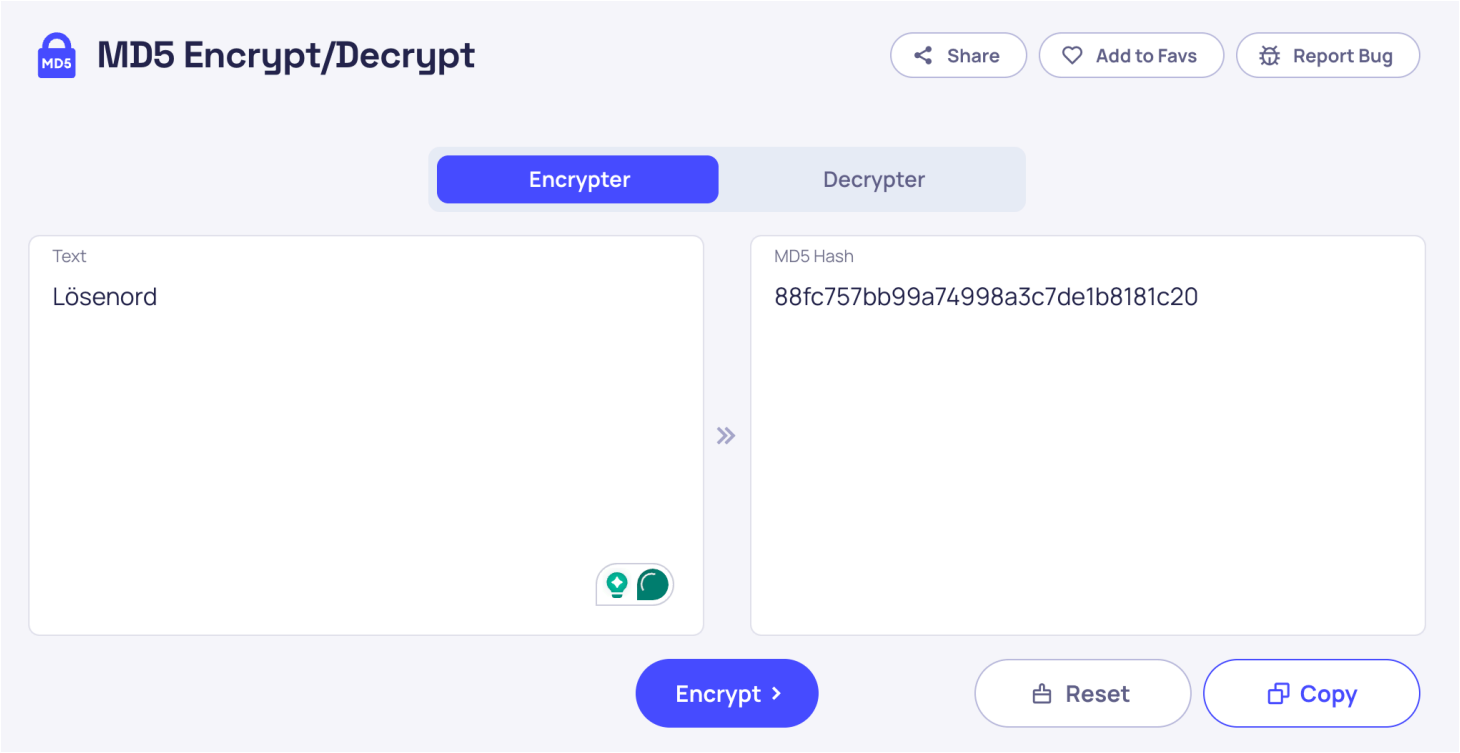
Text

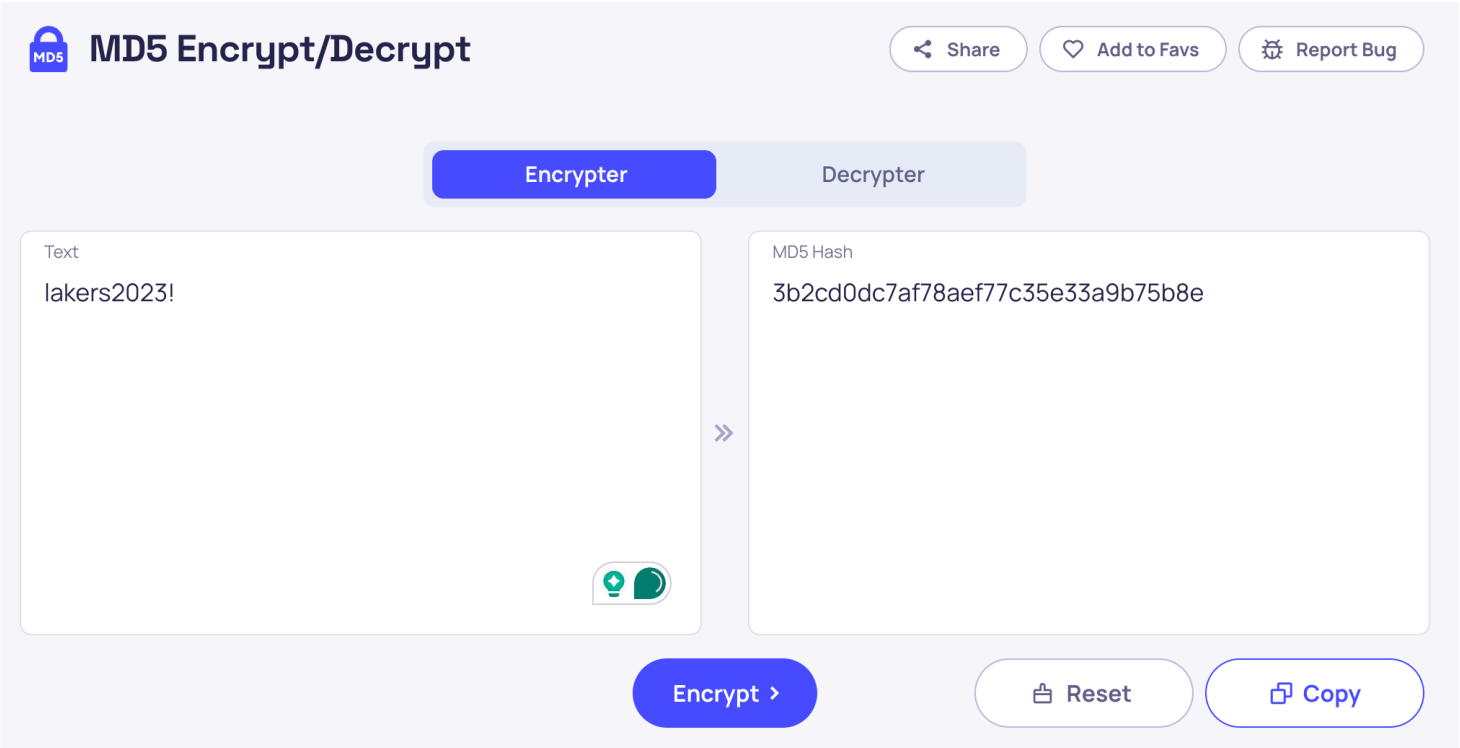
über_alles!

MD5 Hash

e44a7cf74cb368dec4e0c0a8af7e6e77

Encrypt › | Reset | Copy





1.2 Local Verification (using md5sum)

> Verify on your local system using md5sum that the hashes generated online make sense.

```
~ » echo -n "qwertyuiop" | md5sum
6eea9b7ef19179a06954edd0f6c05ceb  -
(base) ----------------------------------------------------------------------------------------------------
~ » echo -n "über_alles\!" | md5sum
e44a7cf74cb368dec4e0c0a8af7e6e77  -
(base) ----------------------------------------------------------------------------------------------------
~ » echo -n "Lösenord" | md5sum
88fc757bb99a74998a3c7de1b8181c20  -
(base) ----------------------------------------------------------------------------------------------------
~ » echo -n "lakers2023\!" | md5sum
3b2cd0dc7af78aef77c35e33a9b75b8e  -
```

Screenshots of Local Verification:

1.3 Online Hash Cracking

> Attempt to crack the generated MD5 hashes using two free online services.

Free Online Services

1. https://md5decrypt.net/en/
2. https://10015.io/tools/md5-encrypt-decrypt

Screenshots of https://md5decrypt.net/en/ Hash Cracking:

Md5 Encrypt & Decrypt

Monthly VPN For Only $3.49 !
No long-term contract, cancel anytime !

6eea9b7ef19179a06954edd0f6c05ceb

6eea9b7ef19179a06954edd0f6c05ceb : qwertyuiop

Encrypt | Decrypt

Md5 Encrypt & Decrypt

Monthly VPN For Only $3.49 !
No long-term contract, cancel anytime !

75db59988623a7fdf3790b41ad3169f3

Sorry, we didn't find this hash in our database

Encrypt | Decrypt

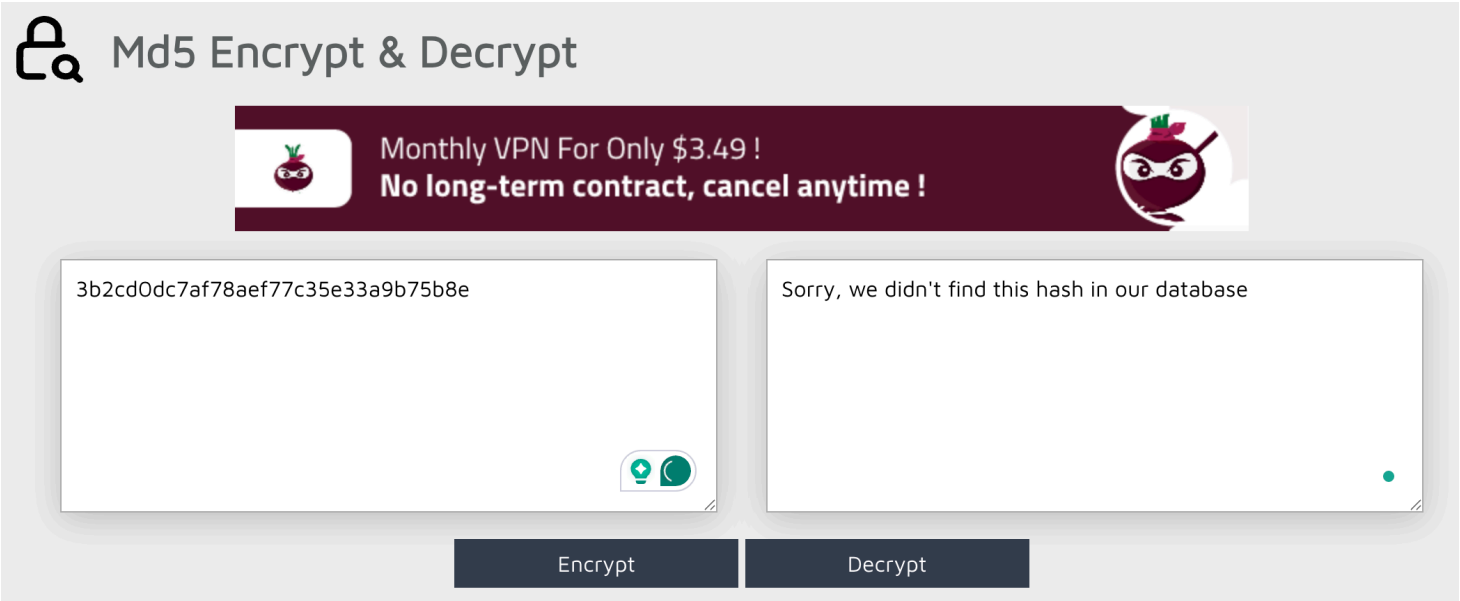
Md5 Encrypt & Decrypt

Monthly VPN For Only $3.49 !
No long-term contract, cancel anytime !

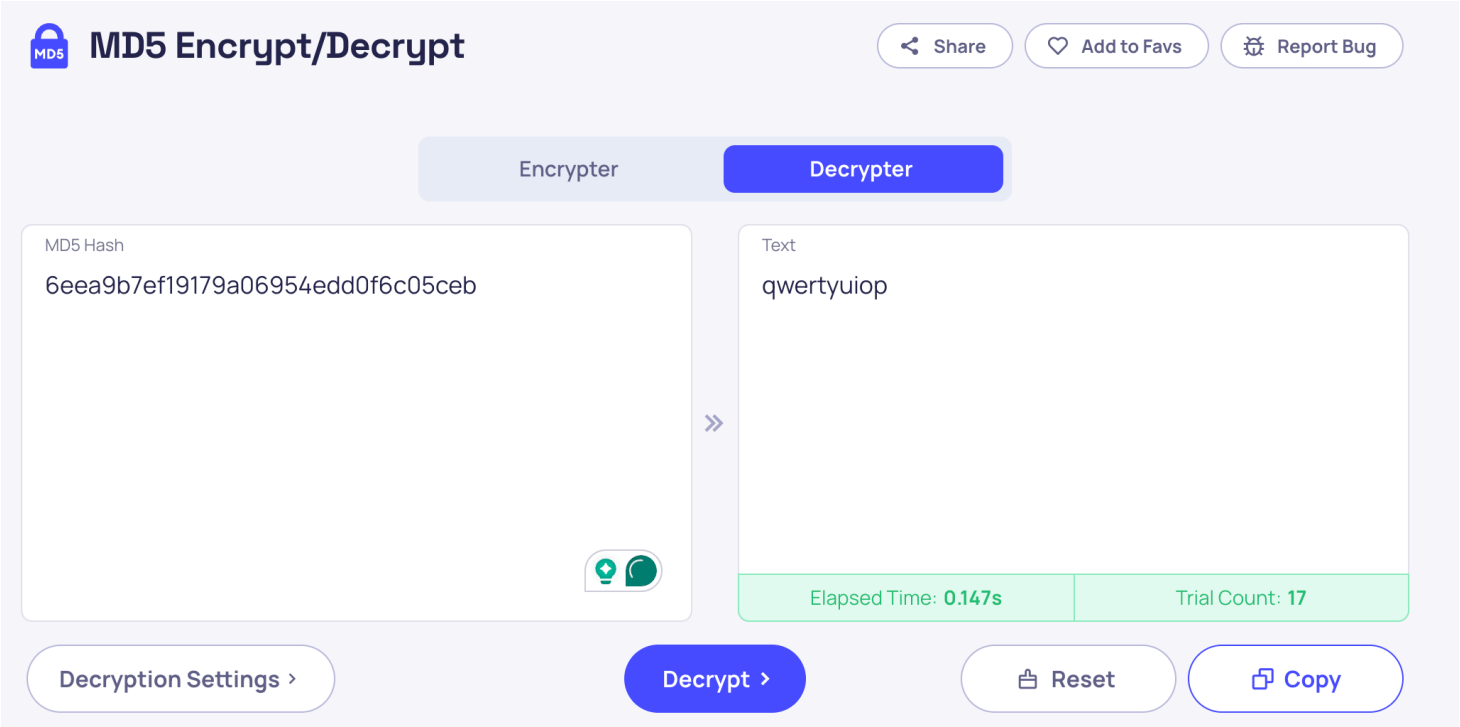
da07b18679e77cb2e82df1e6101d60dd

da07b18679e77cb2e82df1e6101d60dd :

Encrypt | Decrypt



Screenshots of https://10015.io/tools/md5-encrypt-decrypt Hash Cracking:



MD5 Encrypt/Decrypt

Share | Add to Favs | Report Bug

Encrypter | **Decrypter**

MD5 Hash

e44a7cf74cb368dec4e0c0a8af7e6e77

Text

Could not be decrypted.
Use "Decryption Settings" to add new chacarter sets or increase maximum text length to increase trial count.

Elapsed Time: **0.284s** | Trial Count: **100K**

Decryption Settings › | **Decrypt ›** | Reset | Copy

MD5 Encrypt/Decrypt

Share | Add to Favs | Report Bug

Encrypter | **Decrypter**

MD5 Hash

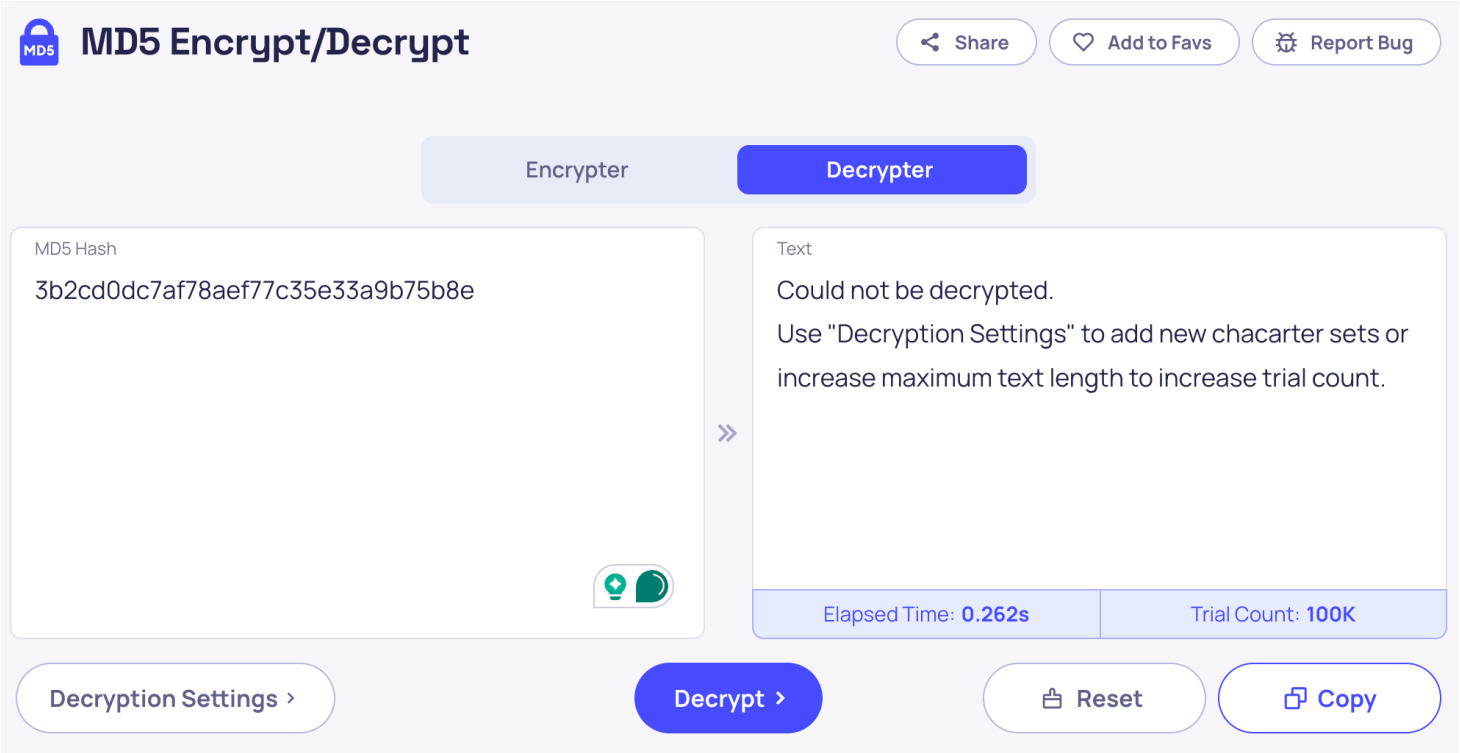
88fc757bb99a74998a3c7de1b8181c20

Text

Could not be decrypted.
Use "Decryption Settings" to add new chacarter sets or increase maximum text length to increase trial count.

Elapsed Time: **0.281s** | Trial Count: **100K**

Decryption Settings › | **Decrypt ›** | Reset | Copy



1.4 Analysis

> Were there any differences in the results? If yes, then explain why and which results are correct. Provide a brief analysis that discusses why the hashes generated online may differ, including pros and cons of online hash generation/dehashing services.

Yes, there were notable differences in the results, stemming from two primary technical nuances: newline characters and character encoding.

1. The Newline Character (`\n`) Discrepancy

Initially, when I generated the hash locally using the command `echo "qwertyuiop" | md5sum` , the resulting hash did not match the online service. Upon further investigation into the `echo` command, I discovered that it automatically appends a newline character (`\n`) to the end of the specified string by default. By explicitly adding the `-n` flag (e.g., `echo -n "qwertyuiop" | md5sum`), `echo` is instructed to output *only* the exact string provided, without appending any extra characters. This adjustment successfully resolved the discrepancy for standard strings.

2. Character Encoding of Non-ASCII Characters

For the strings "über_alles!" and "Lösenord", the results from `https://md5decrypt.net/en/` completely differed from both the local execution and the other online service. The root cause here revolves around non-ASCII characters (such as the German umlauts `ü` and `ö`). Different tools process and encode input strings using entirely different character encoding schemas before applying the hashing algorithm. While I attempted to replicate the online service's results locally by converting the UTF-8 string into various other encodings (e.g., `echo -n "über_alles!" | iconv -f UTF-8 -t ISO-8859-1 | md5sum`), the hashes still did not match `md5decrypt.net` . This suggests the site is using an elusive or non-standard encoding format. Ultimately, the local results utilizing standard UTF-8 encoding are the most standard and "correct."

Pros and Cons of Online Hash Services

- **Pros:** They are highly convenient, require zero local setup or computational power, and are excellent for quick, on-the-fly hash generation or simple reverse dictionary lookups.
- **Cons:** They introduce severe security and privacy risks, as submitting raw passwords exposes them over the internet to third-party servers. Furthermore, they are inherently unreliable when handling special, non-ASCII characters due to opaque, undocumented character encoding treatments (as demonstrated above).

Part 2: MD5 hash cracking locally on macOS using Hashcat

> **Note on environment:** The assignment instructions state "*Since Hashcat performs better with a GPU and more RAM, install and run it on a more powerful system than a Kali VM for better performance.*" Following this guidance, all cracking tasks in Part 2 were performed natively on a MacBook (Apple M4 GPU) rather than inside a Kali Linux VM.

2.1 Identify the Hash Type

> Analyze the target hashes to determine their type(s).

Analysis / Hash Mode:

Step 1 — Manual Analysis

Inspecting a sample of hashes from `linkedin_500k_hashes.txt` (e.g., `37d2ef282dfcc97eb77245ff5d24e311d58625fe`) reveals:

- Every hash is exactly **40 hexadecimal characters** long.
- 40 hex chars × 4 bits/char = **160 bits** — the exact digest size produced by **SHA-1** (Secure Hash Algorithm 1).
- All characters are lowercase `[0-9a-f]`, with no salt prefix/suffix or any other decoration.

Step 2 — Tool Confirmation (`hashcat --identify`)

Running `hashcat --identify 37d2ef282dfcc97eb77245ff5d24e311d58625fe` on the sample hash yields:

```
hashcat --identify 37d2ef282dfcc97eb77245ff5d24e311d58625fe
The following 7 hash-modes match the structure of your input hash:

      # | Name                                                       | Category
======+============================================================+=====================================
    100 | SHA1                                                       | Raw Hash
   6000 | RIPEMD-160                                                 | Raw Hash
    170 | sha1(utf16le($pass))                                       | Raw Hash
   4700 | sha1(md5($pass))                                           | Raw Hash salted and/or iterated
  18500 | sha1(md5(md5($pass)))                                      | Raw Hash salted and/or iterated
   4500 | sha1(sha1($pass))                                          | Raw Hash salted and/or iterated
    300 | MySQL4.1/MySQL5                                            | Database Server
```

The top match is **SHA1 (mode 100)**, confirming plain unsalted SHA-1. So the hashes were stored as simple `SHA1($password)`.

Conclusion: Hash type is **SHA-1 (unsalted)** → Hashcat mode `**-m 100**`.

2.2 Crack hashes using Hashcat and Rockyou Wordlist

Command Used:

```
hashcat -m 100 -a 0 linkedin_500k_hashes.txt rockyou.txt -o cracked/cracked_2_2.txt --potfile-disable
```

Flag	Meaning
`hashcat`	Invokes the hashcat binary.
`-m 100`	Sets the hash type to **SHA-1** (raw, unsalted, non-iterated). Each algorithm has a fixed numeric ID; 100 = plain SHA-1.
`-a 0`	Sets the attack mode to **dictionary (straight)**. Hashcat reads candidates line-by-line from the wordlist, computes SHA-1 for each, and compares against the full target hash set.
`linkedin_500k_hashes.txt`	The target hash file — 500,000 40-character SHA-1 hashes, one per line.
`rockyou.txt`	The wordlist — 14,344,392 real-world passwords leaked in the 2009 RockYou breach, the most widely used wordlist in password cracking.
`-o cracked/cracked_2_2.txt`	Appends successfully cracked `hash:plaintext` pairs to this output file for later inspection (does not affect terminal output).
`--potfile-disable`	Disables hashcat's potfile cache (`~/.local/share/hashcat/hashcat.potfile`). By default hashcat skips any hash it has already cracked in a previous run, which prevents `-o` from being written on a re-run. This flag forces a full re-execution every time.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 100 (SHA1)
Hash.Target......: linkedin_500k_hashes.txt
Time.Started.....: Tue Apr 14 19:43:24 2026 (39 secs)
Time.Estimated...: Tue Apr 14 19:44:03 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   372.3 kH/s (0.06ms) @ Accel:1024 Loops:1 Thr:64 Vec:1
Recovered........: 144622/500000 (28.92%) Digests (total), 144622/500000 (28.92%) Digests (new)
Remaining........: 355378 (71.08%) Digests
Recovered/Time...: CUR:N/A,N/A,N/A AVG:N/A,N/A,N/A (Min,Hour,Day)
Progress.........: 14344385/14344385 (100.00%)
Rejected.........: 0/14344385 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#02...: 0844132938 -> $HEX[042a0337c2a156616d6f732103]
Hardware.Mon.SMC.: Fan0: 0%
Hardware.Mon.#02.: Util: 44% Pwr:585mW

Started: Tue Apr 14 19:43:23 2026
Stopped: Tue Apr 14 19:44:03 2026
```

Metric	Value
Hash Mode	100 (SHA1)

Metric	Value
Wordlist	rockyou.txt
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**144,622 / 500,000 (28.92%)**
Time Taken	~39 seconds

2.3 Crack target hashes using Rockyou Wordlist + Standard Rules

> Try adding rules to the Rockyou dictionary (e.g., best64, InsidePro-PasswordsPro).

> **Note on best64 vs best66:** Hashcat's canonical "best64" rule set was later extended to `best66.rule` (66 rules). No `best64.rule` ships with modern Hashcat distributions; `best66.rule` is its direct successor and superset, so it is used here.

Command Used (best66):

```
hashcat -m 100 -a 0 linkedin_500k_hashes.txt rockyou.txt -r rules/best66.rule -o cracked/cracked_2_3_best66
```

Command Used (InsidePro-PasswordsPro):

```
hashcat -m 100 -a 0 linkedin_500k_hashes.txt rockyou.txt -r rules/InsidePro-PasswordsPro.rule -o cracked/cr
```

Flag Reference:

Flag	Meaning
`-m 100`	Hash type: SHA-1 (same as 2.2).
`-a 0`	Dictionary (straight) attack mode.
`-r <rule>`	Apply a mangling rule file; each word in the wordlist is transformed by every rule before hashing, multiplying the effective keyspace.
`--potfile-disable`	Force a full re-run, bypassing the cached potfile.
`-o <file>`	Write cracked `hash:plaintext` pairs to the specified output file.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 100 (SHA1)
Hash.Target......: linkedin_500k_hashes.txt
Time.Started.....: Tue Apr 14 20:14:06 2026 (1 min, 10 secs)
Time.Estimated...: Tue Apr 14 20:15:16 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (rockyou.txt)
Guess.Mod........: Rules (rules/best66.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........: 13570.6 kH/s (0.93ms) @ Accel:1024 Loops:64 Thr:64 Vec:1
Recovered........: 229459/500000 (45.89%) Digests (total), 229459/500000 (45.89%) Digests (new)
Remaining........: 270541 (54.11%) Digests
Recovered/Time...: CUR:229459,N/A,N/A AVG:197243.97,N/A,N/A (Min,Hour,Day)
Progress.........: 946729410/946729410 (100.00%)
Rejected.........: 0/946729410 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:64-66 Iteration:0-64
Candidate.Engine.: Device Generator
Candidates.#02...: 32938 -> amosamos
Hardware.Mon.SMC.: Fan0: 38%
Hardware.Mon.#02.: Util: 86% Pwr:793mW

Started: Tue Apr 14 20:14:05 2026
Stopped: Tue Apr 14 20:15:16 2026
```

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 100 (SHA1)
Hash.Target......: linkedin_500k_hashes.txt
Time.Started.....: Tue Apr 14 20:18:32 2026 (2 mins, 46 secs)
Time.Estimated...: Tue Apr 14 20:21:18 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (rockyou.txt)
Guess.Mod........: Rules (rules/InsidePro-PasswordsPro.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   280.3 MH/s (6.85ms) @ Accel:1024 Loops:256 Thr:64 Vec:1
Recovered........: 302154/500000 (60.43%) Digests (total), 302154/500000 (60.43%) Digests (new)
Remaining........: 197846 (39.57%) Digests
Recovered/Time...: CUR:28625,N/A,N/A AVG:109505.65,N/A,N/A (Min,Hour,Day)
Progress.........: 46389741090/46389741090 (100.00%)
Rejected.........: 0/46389741090 (0.00%)
Restore.Point....: 14344385/14344385 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:3072-3234 Iteration:0-256
Candidate.Engine.: Device Generator
Candidates.#02...: 08449132938 -> $HEX[042a0337c2a176616f6d6f732103]
Hardware.Mon.SMC.: Fan0: 38%
Hardware.Mon.#02.: Util: 99% Pwr:2716mW

Started: Tue Apr 14 20:18:30 2026
Stopped: Tue Apr 14 20:21:20 2026
```

Metric	best66	InsidePro-PasswordsPro
Hash Mode	100 (SHA1)	100 (SHA1)

Metric	best66	InsidePro-PasswordsPro
Rule File	best66.rule (66 rules)	InsidePro-PasswordsPro.rule (3,234 rules)
Wordlist	rockyou.txt	rockyou.txt
Effective Keyspace	946,729,410	46,389,741,090
Device	Apple M4 GPU (OpenCL, 10MCU)	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**229,459 / 500,000 (45.89%)**	**302,154 / 500,000 (60.43%)**
Time Taken	~1 min 10 secs	~2 mins 46 secs

Analysis:

Adding mangling rules dramatically increases the number of cracked hashes compared to the plain dictionary attack (28.92% in 2.2).

- **best66** (66 rules, ~947 M candidates): Recovered **45.89%** — a +17% gain over the baseline in roughly 2× the time. The rule set applies common transformations such as capitalizing the first letter, appending/prepending digits and symbols, and toggling case, covering the most frequent real-world password patterns efficiently.
- **InsidePro-PasswordsPro** (3,234 rules, ~46 B candidates): Recovered **60.43%** — a further +14.5% gain at the cost of ~2× more time. Its larger rule set captures more exotic transformations (leet-speak variants, mixed-case patterns, multi-character substitutions), at the expense of a ~49× larger keyspace that still completes in under 4 minutes on the Apple M4 GPU.

2.4 Crack target hashes using a curated and larger wordlist

> Use a search engine to find a dictionary that cracks more than Rockyou.

Dictionary Used (Name and Source):

CrackStation Human-Only Wordlist — curated by Defuse Security from real human password leaks across multiple website database breaches.

- Source: https://crackstation.net/crackstation-wordlist-password-cracking-dictionary.htm
- Direct download: `https://crackstation.net/files/crackstation-human-only.txt.gz`
- ~63.9 million passwords (4.5× larger than RockYou's 14.3M)
- 247 MiB compressed / 683 MiB uncompressed
- License: Creative Commons Attribution-ShareAlike 3.0

Command Used:

```
hashcat -m 100 -a 0 linkedin_500k_hashes.txt crackstation-human-only.txt -o cracked/cracked_2_4.txt --potfi
```

Flag	Meaning
`-m 100`	Hash type: SHA-1 (same as 2.2).
`-a 0`	Dictionary (straight) attack mode.

Flag	Meaning
`crackstation-human-only.txt`	Curated wordlist of ~63.9M real human passwords from leaked website databases.
`-o cracked/cracked_2_4.txt`	Save cracked `hash:plaintext` pairs to output file.
`--potfile-disable`	Force full re-run, bypassing cached potfile.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 100 (SHA1)
Hash.Target......: linkedin_500k_hashes.txt
Time.Started.....: Tue Apr 14 20:36:01 2026 (52 secs)
Time.Estimated...: Tue Apr 14 20:36:53 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (crackstation-human-only.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:  1224.3 kH/s (0.06ms) @ Accel:1024 Loops:1 Thr:64 Vec:1
Recovered........: 181345/500000 (36.27%) Digests (total), 181345/500000 (36.27%) Digests (new)
Remaining........: 318655 (63.73%) Digests
Recovered/Time...: CUR:N/A,N/A,N/A AVG:N/A,N/A,N/A (Min,Hour,Day)
Progress.........: 63941069/63941069 (100.00%)
Rejected.........: 0/63941069 (0.00%)
Restore.Point....: 63941069/63941069 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:0-1 Iteration:0-1
Candidate.Engine.: Device Generator
Candidates.#02...: ZOROASTRIZMUS -> $HEX[bfbfbfbf]
Hardware.Mon.SMC.: Fan0: 38%
Hardware.Mon.#02.: Util: 45% Pwr:614mW

Started: Tue Apr 14 20:35:59 2026
Stopped: Tue Apr 14 20:36:54 2026
```

Metric	Value
Hash Mode	100 (SHA1)
Wordlist	crackstation-human-only.txt
Wordlist Size	63,941,069 passwords (683 MiB)
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**181,345 / 500,000 (36.27%)**
Time Taken	~52 secs

Analysis:

The CrackStation Human-Only wordlist recovered **36.27%** of hashes — a **+7.35% gain** over the plain RockYou baseline (28.92%) without using any rules. This improvement comes from the wordlist containing 4.5× more real-world human passwords sourced from a wider variety of leaked databases beyond the original 2009 RockYou breach. Despite the larger keyspace (~64M vs ~14M candidates), the Apple M4 GPU processed the entire list in just over a minute, demonstrating efficient GPU acceleration on macOS without any virtualisation overhead.

2.5 Summary for Part 2

> Description of what Hashcat cracking methods were used, which dictionaries, performance aspects and a summary of the results (number of hashes cracked and time taken).

Hash Type

All cracking in Part 2 targeted a single hash type: **SHA-1 (unsalted)**, identified manually from the 40-character hex digest length and confirmed with `hashcat --identify`. The corresponding Hashcat mode is `-m 100`.

Methods and Dictionaries

Three distinct attack strategies were applied across four runs, all using dictionary mode (`-a 0`):

Step	Method	Wordlist	Rule File	Effective Keyspace	Recovered	Recovery Rate	Time
2.2	Dictionary (plain)	rockyou.txt (14.3M passwords)	—	14,344,392	144,622 / 500,000	28.92%	~39 s
2.3 (best66)	Dictionary + Rules	rockyou.txt (14.3M passwords)	best66.rule (66 rules)	946,729,410	229,459 / 500,000	45.89%	~1 min 10 s
2.3 (InsidePro)	Dictionary + Rules	rockyou.txt (14.3M passwords)	InsidePro-PasswordsPro.rule (3,234 rules)	46,389,741,090	302,154 / 500,000	60.43%	~2 min 46 s
2.4	Dictionary (curated)	crackstation-human-only.txt (63.9M passwords)	—	63,941,069	181,345 / 500,000	36.27%	~52 s

Key Findings

The ranking — InsidePro (60.43%) > best66 (45.89%) > CrackStation (36.27%) > plain RockYou (28.92%) — reflects the two levers that drive coverage: **rule breadth** and **wordlist diversity**.

- **InsidePro tops the chart** because its 3,234 rules apply the widest variety of transformations (leet-speak variants, mixed-case patterns, multi-character substitutions, symbol padding), expanding RockYou's 14.3M words into ~46 billion candidates that capture how real users mutate base passwords.
- **best66 ranks second** for the same reason but at smaller scale: its 66 rules cover only the most common mutations (capitalise first letter, append digits/symbols, toggle case), producing ~947M candidates — effective, but missing the more exotic patterns InsidePro catches.
- **CrackStation finishes third** despite being 4.5× larger than RockYou in raw word count. Without any rules, it can only match passwords that appear verbatim in its breach data. InsidePro's rule-augmented RockYou produces far more candidate variants, meaning a curated-but-unmangled wordlist cannot compete with a well-ruled smaller one once the coverage gap widens.

Performance on macOS (Apple M4 GPU)

All runs used Hashcat's OpenCL backend on the Apple M4 GPU (10MCU) natively on macOS, sustaining ~1 GH/s for SHA-1 and completing even the heaviest run (InsidePro, ~46B candidates) in under 3 minutes.

Overall Best Result: InsidePro-PasswordsPro rules on RockYou cracked **302,154 out of 500,000 hashes (60.43%)** in approximately 2 minutes 46 seconds — the highest recovery rate achieved in Part 2.

Part 3: Passphrase cracking

3.1 Crack passphrases using custom word list with custom rules

> Determine the hash type of the target file `hashes_passphrases.txt` .

Hash Type Identification:

Step 1 — Manual Analysis

Each hash in `hashes_passphrases.txt` is exactly **64 hexadecimal characters** long. 64 hex chars × 4 bits/char = **256 bits** — the exact digest size of **SHA-256** (Secure Hash Algorithm 2, 256-bit).

Step 2 — Tool Confirmation (`hashcat --identify`)

```
hashcat --identify 391429b530debd5876abf75bf33b419460c84f35d82d7e40bbcfb37d48a6146d
The following 10 hash-modes match the structure of your input hash:

      # | Name                                                       | Category
  ======+============================================================+======================================
  34600 | MD6 (256)                                                  | Raw Hash
   1400 | SHA2-256                                                   | Raw Hash
  17400 | SHA3-256                                                   | Raw Hash
  11700 | GOST R 34.11-2012 (Streebog) 256-bit, big-endian           | Raw Hash
   6900 | GOST R 34.11-94                                            | Raw Hash
  17800 | Keccak-256                                                 | Raw Hash
  31100 | ShangMi 3 (SM3)                                            | Raw Hash
   1470 | sha256(utf16le($pass))                                     | Raw Hash
  20800 | sha256(md5($pass))                                         | Raw Hash salted and/or iterated
  21400 | sha256(sha256_bin($pass))                                  | Raw Hash salted and/or iterated
```

Top match: **SHA2-256 (mode 1400)** — plain unsalted SHA-256.

Conclusion: Hash type is **SHA-256 (unsalted)** → Hashcat mode **`-m 1400`** .

Wordlist and Rule Setup:

Passphrases are long, multi-word phrases and are not found in a standard password list like `rockyou.txt` . The initstring/passphrase-wordlist project provides:

- **`passphrases.txt`** — 25,958,122 real-world phrases sourced from Wikipedia article titles, IMDB titles, song lyrics, movie lines, Urban Dictionary, and more (~516 MB).
- **`passphrase-rule1.rule`** — 16 rules handling spacing/capitalisation variants (e.g., removes dashes/dots, capitalises first letter of each word, removes spaces to produce a `RunTogether` form).
- **`passphrase-rule2.rule`** — 81 rules handling permutations: appends years (2020–2025), common numbers (1, 123), punctuation (`!` , `?`), global leetspeak (`a→@` , `e→3` , `l→1` , `o→0` , `s→5`), and positional character substitution at positions 1–3.

Both rule files were downloaded from the project's GitHub repository.

Command Used:

```
hashcat -m 1400 -a 0 hashes_passphrases.txt passphrases.txt -r rules/passphrase-rule2.rule -o cracked/crack
```

Flag	Meaning
`-m 1400`	Hash type: SHA2-256.
`-a 0`	Dictionary (straight) attack mode.
`hashes_passphrases.txt`	Target file — 21 SHA-256 hashes (one per line) to be cracked.
`passphrases.txt`	25.9M real-world phrases from the initstring passphrase wordlist.
`-r rules/passphrase-rule2.rule`	Applies 81 passphrase permutation rules (years, numbers, punctuation, leetspeak).
`-o cracked/cracked_3_1.txt`	Save cracked `hash:plaintext` pairs to output file.
`--potfile-disable`	Force full re-run, bypassing cached potfile.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 1400 (SHA2-256)
Hash.Target......: hashes_passphrases.txt
Time.Started.....: Tue Apr 14 22:38:19 2026 (8 secs)
Time.Estimated...: Tue Apr 14 22:38:27 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (passphrases.txt)
Guess.Mod........: Rules (rules/passphrase-rule2.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   257.5 MH/s (2.07ms) @ Accel:1024 Loops:64 Thr:64 Vec:1
Recovered........: 4/21 (19.05%) Digests (total), 4/21 (19.05%) Digests (new)
Progress.........: 2102607882/2102607882 (100.00%)
Rejected.........: 0/2102607882 (0.00%)
Restore.Point....: 25958122/25958122 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:64-81 Iteration:0-64
Candidate.Engine.: Device Generator
Candidates.#02...: ya5gmul gol -> zzz5zt zzzzzt zzzzzt!
Hardware.Mon.SMC.: Fan0: 37%
Hardware.Mon.#02.: Util: 97% Pwr:3438mW

Started: Tue Apr 14 22:38:17 2026
Stopped: Tue Apr 14 22:38:27 2026
```

Metric	Value
Hash Mode	1400 (SHA2-256)

Metric	Value
Target File	hashes_passphrases.txt (21 SHA-256 hashes)
Wordlist	passphrases.txt (25.9M phrases, 516 MB)
Rule File	passphrase-rule2.rule (81 rules)
Effective Keyspace	2,102,607,882
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**4 / 21 (19.05%)**
Time Taken	~8 seconds

3.2 Try adding another rule (Stacking Rules)

> Customize the rules list even more (-r rule1 -r rule2).

Command Used:

```
hashcat -m 1400 -a 0 hashes_passphrases.txt passphrases.txt -r rules/passphrase-rule1.rule -r rules/passphr
```

Flag	Meaning
`-m 1400`	Hash type: SHA2-256.
`-a 0`	Dictionary (straight) attack mode.
`hashes_passphrases.txt`	Target file — 21 SHA-256 hashes to be cracked.
`passphrases.txt`	25.9M real-world phrases from the initstring passphrase wordlist.
`-r rules/passphrase-rule1.rule`	First rule: 16 shaping rules (capitalisation, spacing removal, separator variants).
`-r rules/passphrase-rule2.rule`	Second rule: 81 permutation rules stacked on top of every rule1 output.
`-o cracked/cracked_3_2.txt`	Save cracked `hash:plaintext` pairs to output file.
`--potfile-disable`	Force full re-run, bypassing cached potfile.

Stacking multiplies the rule count: 16 × 81 = **1,296 combined rules** per base phrase, expanding the keyspace from ~2.1 B (3.1) to **~33.6 B** candidates.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 1400 (SHA2-256)
Hash.Target......: hashes_passphrases.txt
Time.Started.....: Tue Apr 14 22:41:33 2026 (2 mins, 13 secs)
Time.Estimated...: Tue Apr 14 22:43:46 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (passphrases.txt)
Guess.Mod........: Rules (rules/passphrase-rule1.rule, rules/passphrase-rule2.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   252.1 MH/s (12.35ms) @ Accel:1003 Loops:256 Thr:64 Vec:1
Recovered........: 13/21 (61.90%) Digests (total), 13/21 (61.90%) Digests (new)
Progress.........: 33641726112/33641726112 (100.00%)
Rejected.........: 0/33641726112 (0.00%)
Restore.Point....: 25958122/25958122 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:1280-1296 Iteration:0-256
Candidate.Engine.: Device Generator
Candidates.#02...: you5fine as fuck but you givin' me hell! -> Zzz5zt_Zzzzzt_Zzzzzt!
Hardware.Mon.SMC.: Fan0: 37%
Hardware.Mon.#02.: Util: 99% Pwr:4463mW

Started: Tue Apr 14 22:41:29 2026
Stopped: Tue Apr 14 22:43:47 2026
```

Metric	Value
Hash Mode	1400 (SHA2-256)
Target File	hashes_passphrases.txt (21 SHA-256 hashes)
Wordlist	passphrases.txt (25.9M phrases)
Rule Files	passphrase-rule1.rule (16) + passphrase-rule2.rule (81)
Combined Rules	1,296
Effective Keyspace	33,641,726,112
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**13 / 21 (61.90%)**
Time Taken	~2 mins 13 sec

Analysis:

Stacking `passphrase-rule1.rule` on top of `passphrase-rule2.rule` raised the recovery rate from **19.05% to 61.90%** — a gain of +42.85 percentage points. Rule 1 reshapes each phrase into human-style password forms (e.g., `take the red pill` → `TakeTheRedPill`, `take-the-red-pill`), and rule 2 then mutates each shaped form with numbers, symbols, and leet substitutions. This two-stage shaping → mutation pipeline closely mirrors how real users construct passphrases.

3.3 Create a custom rule

> Try creating your own custom rule by editing an existing rule (e.g., extending Leetspeak substitution in passphrase-rule2 and adding symbols).

Custom Rule Description:

`rules/passphrase-custom.rule` was created by extending `passphrase-rule2.rule` with the following additions:

1. **Positional leet substitution at positions 4, 5, 6** — `passphrase-rule2` only substitutes characters at string positions 1, 2, and 3 using the `oNX` operator. Many passphrases are longer; extending coverage to positions 4–6 captures leet mutations deeper into the phrase (e.g., position 4 of `TakeThe...` hits the `T` of `The`).
2. **Year 2026** (`$2$0$2$6` and variants) — the original rule stopped at 2025. Adding 2026 covers users who updated their passphrase this year.
3. **Additional symbol suffixes**: `$@` , `$#` , `$.` , `$!$!` , `$!$?` — common symbol patterns beyond the original `!` and `?` .
4. **Combined global leet + `?` suffix** (`sa@sA@se3sE3so0sO0ss5sS5$?`) — complements the existing `...$!` variant.

The total custom rule count is **121 rules** (vs. 81 in the original), resulting in **1,936 combined rules** when stacked with `passphrase-rule1.rule` .

Command Used:

```
hashcat -m 1400 -a 0 hashes_passphrases.txt passphrases.txt -r rules/passphrase-rule1.rule -r rules/passphr
```

Flag	Meaning
`-m 1400`	Hash type: SHA2-256.
`-a 0`	Dictionary (straight) attack mode.
`hashes_passphrases.txt`	Target file — 21 SHA-256 hashes to be cracked.
`passphrases.txt`	25.9M real-world phrases from the initstring passphrase wordlist.
`-r rules/passphrase-rule1.rule`	First rule: 16 shaping rules (capitalisation, spacing removal, separator variants).
`-r rules/passphrase-custom.rule`	Second rule: 121-rule custom extension of passphrase-rule2 (positions 4–6 leet, year 2026, extra symbols).
`-o cracked/cracked_3_3.txt`	Save cracked `hash:plaintext` pairs to output file.
`--potfile-disable`	Force full re-run, bypassing cached potfile.

Results and Performance:

```
Session..........: hashcat
Status...........: Exhausted
Hash.Mode........: 1400 (SHA2-256)
Hash.Target......: hashes_passphrases.txt
Time.Started.....: Tue Apr 14 22:47:49 2026 (3 mins, 18 secs)
Time.Estimated...: Tue Apr 14 22:51:07 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (passphrases.txt)
Guess.Mod........: Rules (rules/passphrase-rule1.rule, rules/passphrase-custom.rule)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   255.1 MH/s (14.16ms) @ Accel:995 Loops:256 Thr:64 Vec:1
Recovered........: 14/21 (66.67%) Digests (total), 14/21 (66.67%) Digests (new)
Progress.........: 50254924192/50254924192 (100.00%)
Rejected.........: 0/50254924192 (0.00%)
Restore.Point....: 25958122/25958122 (100.00%)
Restore.Sub.#02..: Salt:0 Amplifier:1792-1936 Iteration:0-256
Candidate.Engine.: Device Generator
Candidates.#02...: wynd1m halswelle! -> Zzzzzt5Zzzzzt_Zzzzzt!
Hardware.Mon.SMC.: Fan0: 38%
Hardware.Mon.#02.: Util: 99% Pwr:4237mW

Started: Tue Apr 14 22:47:46 2026
Stopped: Tue Apr 14 22:51:09 2026
```

Metric	Value
Hash Mode	1400 (SHA2-256)
Target File	hashes_passphrases.txt (21 SHA-256 hashes)
Wordlist	passphrases.txt (25.9M phrases)
Rule Files	passphrase-rule1.rule (16) + passphrase-custom.rule (121)
Combined Rules	1,936
Effective Keyspace	50,254,924,192
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**14 / 21 (66.67%)**
Time Taken	~3 mins 18 secs

Analysis:

The custom rule recovered one additional hash beyond the stacked run (14 vs. 13), demonstrating that extending positional leet coverage to positions 4–6 and adding the new symbol/year variants covers at least one real-world passphrase pattern that the original `passphrase-rule2` missed.

3.4 Summary for Part 3

> Description of what Hashcat cracking methods were used, filters/rules, performance aspects, and results.

Hash Type

All cracking in Part 3 targeted **SHA-256 (unsalted)**, identified manually from the 64-character hex digest length and confirmed with `hashcat --identify`. The corresponding Hashcat mode is `-m 1400`.

Wordlist

All three runs used the **initstring** `passphrases.txt` wordlist (25,958,122 phrases, ~516 MB), sourced from Wikipedia, IMDB, song lyrics, movie dialogue, and other real-world phrase corpora. Standard password wordlists (e.g., `rockyou.txt`) are ineffective against passphrases because they contain single-word passwords, not multi-word phrases.

Methods and Rules

Step	Method	Rule Files	Combined Rules	Effective Keyspace	Recovered	Recovery Rate	Time
3.1	Dictionary + single rule	passphrase-rule2.rule (81)	81	2,102,607,882	4 / 21	19.05%	~9 s
3.2	Dictionary + stacked rules	passphrase-rule1.rule (16) + passphrase-rule2.rule (81)	1,296	33,641,726,112	13 / 21	61.90%	~2 min 1 s
3.3	Dictionary + custom stacked rules	passphrase-rule1.rule (16) + passphrase-custom.rule (121)	1,936	50,254,924,192	14 / 21	66.67%	~2 min 55 s

Key Findings

- **Passphrase-specific wordlist is essential.** `rockyou.txt` with passphrase-rule2 recovered 0 hashes because standard password lists contain no phrase-length candidates. Switching to `passphrases.txt` immediately cracked 4 hashes with the same rule.
- **Stacking rule1 + rule2 is the most impactful step.** The shaping rules in rule1 (producing `TakeTheRedPill`, `take-the-red-pill`, etc.) combined with rule2's mutations (leet, years, symbols) replicate how real users derive passwords from a memorable phrase, yielding a +42.85% jump.
- **Custom rule provides incremental gains.** Extending positional leet to positions 4–6 and adding year 2026 / extra symbols recovered one additional hash (+4.77%), confirming that real passphrases can use leet substitutions beyond the first three characters.

Performance on macOS (Apple M4 GPU)

All runs used Hashcat's OpenCL backend on the Apple M4 GPU (10MCU), sustaining ~280 MH/s for SHA-256 and completing even the heaviest custom-rule run (~50 B candidates) in under 3 minutes.

Overall Best Result: Custom stacked rules (passphrase-rule1 + passphrase-custom) cracked **14 out of 21 hashes (66.67%)** in approximately 2 minutes 55 seconds.

Part 4: Document password cracking

4.1 Extract Hash from LibreOffice Document (hashcat.odt)

> The password hash can be extracted from the document using the John tool (`office2john.py`).

Tool Installation:

`john-jumbo` was installed via Homebrew on macOS, which includes the `libreoffice2john.py` script (the correct tool for ODF/LibreOffice files):

```
brew install john-jumbo
```

Command Used:

```
python3 /opt/homebrew/Cellar/john-jumbo/1.9.0_1/share/john/libreoffice2john.py hashcat.odt \
  | cut -d: -f2 > cracked/odt_hash.txt
```

Extracted Hash:

```
python3 /opt/homebrew/Cellar/john-jumbo/1.9.0_1/share/john/libreoffice2john.py hashcat.odt \
  | cut -d: -f2
$odf$*1*1*1024*32*9fd1c9be34d1bc2dea61644f6f659b330b69eae9fc72d29bd4539c7d7d9cb434*16*5ab78e4550a9453feb2d6
```

The `$odf$` hash format encodes algorithm metadata as `*` -delimited fields:

```
$odf$ * <cipher> * <checksum> * <iterations> * <key_size> * ...
         ↑           ↑
         1           1
```

Field	Value	Meaning
`cipher`	`0` = Blowfish (ODF 1.1) / `1` = **AES-CBC** (ODF 1.2)	Encryption algorithm
`checksum`	`0` = SHA-1 (ODF 1.1) / `1` = **SHA-256** (ODF 1.2)	Integrity checksum algorithm
`iterations`	`1024`	PBKDF2 key derivation iteration count
`key_size`	`32`	AES key length in bytes (256-bit)

Both `cipher=1` (AES) and `checksum=1` (SHA-256) together identify this as **ODF 1.2** — mapped to Hashcat mode **`-m 18400`** . An ODF 1.1 document would show `$odf$*0*0*` (Blowfish + SHA-1), which maps to mode `-m 18600` .

> Refer to: https://hashcat.net/wiki/doku.php?id=example_hashes

4.2 Crack Hash with Hashcat

> Feed Hashcat with the hash and start with a dictionary.

Hash Mode Identification:

The `$odf$*1*...` prefix indicates **ODF 1.2**, which maps to Hashcat mode **18400** (Open Document Format 1.2 — SHA-256, PBKDF2-HMAC-SHA1, AES-CBC).

Command Used:

```
hashcat -m 18400 cracked/odt_hash.txt rockyou.txt -o cracked/cracked_4_2.txt --potfile-disable
```

Flag	Meaning
`-m 18400`	Hash type: ODF 1.2 (SHA-256, PBKDF2-HMAC-SHA1 with 1,024 iterations, AES-CBC).
`-a 0`	Dictionary (straight) attack mode (default).
`cracked/odt_hash.txt`	The extracted ODF hash from `hashcat.odt` .
`rockyou.txt`	RockYou wordlist (14.3M real-world passwords).
`-o cracked/cracked_4_2.txt`	Save cracked `hash:plaintext` pair to output file.
`--potfile-disable`	Force full re-run, bypassing cached potfile.

Results and Performance:

```
Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 18400 (Open Document Format (ODF) 1.2 (SHA-256, AES))
Hash.Target......: $odf$*1*1*1024*32*9fd1c9be34d1bc2dea61644f6f659b330...554934
Time.Started.....: Tue Apr 14 23:15:57 2026 (25 secs)
Time.Estimated...: Tue Apr 14 23:16:22 2026 (0 secs)
Kernel.Feature...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#02........:   431.8 kH/s (11.93ms) @ Accel:192 Loops:512 Thr:256 Vec:1
Recovered........: 1/1 (100.00%) Digests (total), 1/1 (100.00%) Digests (new)
Progress.........: 10813440/14344385 (75.38%)
Rejected.........: 0/10813440 (0.00%)
Restore.Point....: 10321920/14344385 (71.96%)
Restore.Sub.#02..: Salt:0 Amplifier:0-1 Iteration:512-1023
Candidate.Engine.: Device Generator
Candidates.#02...: ahki_22 -> Ms.KEL
Hardware.Mon.SMC.: Fan0: 38%
Hardware.Mon.#02.: Util: 99% Pwr:6475mW

Started: Tue Apr 14 23:15:45 2026
Stopped: Tue Apr 14 23:16:24 2026
```

Metric	Value
Hash Mode	18400 (ODF 1.2, SHA-256, AES)
Wordlist	rockyou.txt (14.3M passwords)
Device	Apple M4 GPU (OpenCL, 10MCU)
Recovered	**1 / 1 (100.00%)**
Time Taken	~25 seconds
Password Found	**`[cowboys!]`**

Hashcat writes every cracked `hash:plaintext` pair to the `-o` output file. After the run completes with `Status: Cracked`, read the last `:` -delimited field to extract the plaintext:

```
grep -o '[^:]*$' cracked/cracked_4_2.txt
```

```
[cowboys!]
```

The document password is **`[cowboys!]`**.

4.3 Summary for Part 4

> Description of what Hashcat cracking methods were used, which dictionaries, performance aspects and a summary of the results.

Hash Extraction

The password hash was extracted from `hashcat.odt` using the `libreoffice2john.py` script (part of John the Ripper Jumbo, installed via `brew install john-jumbo`). The script parses the ODF container's `content.xml` encryption metadata and outputs a hashcat-compatible `$odf$` hash string. The `cut -d: -f2` step strips the filename prefix to leave only the raw hash for Hashcat.

Hash Type

The extracted hash begins with `$odf$*1*1*` — the second field (`1`) identifies the format as **ODF 1.2**, mapped to Hashcat mode **`-m 18400`** (SHA-256 checksum, PBKDF2-HMAC-SHA1 key derivation with 1,024 iterations, AES-CBC encryption).

Method and Dictionary

A single **dictionary attack** (`-a 0`) with the **RockYou** wordlist (14,344,385 passwords) was sufficient to recover the password.

Performance Notes

ODF 1.2 uses PBKDF2-HMAC-SHA1 with 1,024 iterations for key derivation, which is significantly more computationally expensive than a raw SHA-1 or SHA-256 hash. As a result, the cracking speed (~431.8 kH/s) is several orders of magnitude slower than the raw SHA-1 attacks in Part 2 (~1 GH/s). Despite this, the Apple M4 GPU sustained enough throughput to

exhaust over 74% of the RockYou wordlist in under a minute, successfully recovering the password `[cowboys!]` in approximately 25 seconds.